

741. for which integers u are there solutions to $x^2 - y^2 = u$?

$$x^2 - y^2 = u \quad \text{means} \quad \underbrace{(x+y)}_{\substack{|| \\ b}} \underbrace{(x-y)}_{\substack{|| \\ a}} = u$$

$$\text{with } x = (a+b)/2 \\ y = (b-a)/2$$

$$u = (\text{even} \times \text{even}) \iff 4|u \\ \text{or } (\text{odd} \times \text{odd})$$

$$(x, y) \longleftrightarrow (a, b) \\ \text{bijection with same parity}$$

$$\text{claim } 4|u \text{ iff } a = \text{even} \\ b = \text{even}$$

$$\text{claim } u \equiv 1 \pmod{2} \text{ iff } a = \text{odd} \\ b = \text{odd}$$

$$\Rightarrow u = 2 \times \left(\frac{u}{2}\right)$$

$$\Rightarrow u = 1 \times u$$

$$\iff 4 \mid 2a' \times 2b' \\ \quad \quad \quad || \quad \quad || \\ \quad \quad \quad a \quad \quad b$$

$$\iff \text{odd} \times \text{odd} = \text{odd} \equiv 1 \pmod{2}$$

1041. $x^2 + y^2 = z^4$ has infinitely many solutions
with $(x, y, z) = 1$

∴ primitive $x^2 + y^2 = z^2$

$$\text{then } x = v^2 - s^2 \quad \text{with } ① v > s > 0$$

$$y = 2vs \quad ② (v, s) = 1$$

$$z = v^2 + s^2 \quad ③ v, s \text{ opposite parity}$$

$$\text{let } v = t^2 - 1 \quad \text{with } ① t \geq 4 \quad \text{i.e. } t^2 - 1 \geq 2t$$

$$s = 2t \quad ② t \equiv 2 \pmod{4} \quad v = (t+1)(t-1)$$

$$z = 2t$$

$$\text{then } z = \underbrace{(t^2 + 1)^2}_{||}$$

||

증명되어
찾던 z^2 값!

$$(t+1, t-1) = 1 \quad (v, s) = 1$$

$$(t+1, t) = 1$$

$$(t-1, t) = 1$$

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1141. $x^2 + y^2 = 2z^2$

$\Rightarrow x^2$ 과 y^2 이 홀수이면 z^2 가 홀수여야 한다.

case ① x, y odd

$$x^2 + y^2 = 2z^2$$

$$\begin{cases} r = (x+y)/2 \rightarrow \\ s = (x-y)/2 \rightarrow \end{cases}$$

then $z^2 = r^2 + s^2$ $\therefore (r+s)^2 + (r-s)^2 = 2(r^2 + s^2)$

• $r = m^2 - n^2$

• $s = 2mn$

• $z = m^2 + n^2$

$\Rightarrow x = m^2 + 2mn - n^2$

$y = m^2 - 2mn - n^2$

primitive
solutions

$x = k(m^2 + 2mn - n^2)$

$y = k(m^2 - 2mn - n^2)$

$z = k(m^2 + n^2)$

case ② x, y even

$x_0^2 + y_0^2 = 2z_0^2$ $x_0 = 2x_1$

$\downarrow$ $y_0 = 2y_1$

$x_1^2 + y_1^2 = 2z_1^2$ $z_0 = 2z_1$

$\downarrow$ 반복 | 계속된다.

$\rightarrow \leftarrow$

1. $x^2 + y^2 = 9z + 3$ has no integral solution.

$0 \pm 1 \pm 2 \pm 3 \pm 4$

$\downarrow \downarrow \downarrow \downarrow \downarrow$

$0 \quad 1 \quad 4 \quad 0 \quad -2$

$0, 1, 4, 7$

$x^2 + y^2 \equiv 3 \pmod{9}$

(+) $0 \quad 1 \quad 4 \quad 7$ (-2)

$0 \quad 0 \quad 1 \quad 4 \quad 7$

$1 \quad 1 \quad 2 \quad 5 \quad 8$

$4 \quad 4 \quad 5 \quad 8 \quad 2$

$(-2) \quad 7 \quad 7 \quad 8 \quad 2 \quad 5$

$\leftarrow$ 3이
없음.

2. $x^2 + 2y^2 = 8z + 5$ has no integral solution



$$x^2 + 2y^2 \equiv 5 \pmod{8}$$

arbitrary $x^2 \equiv 0, 1, 4 \pmod{8}$

$$\begin{array}{ccc} & x^2 & \\ \hline 0 & 1 & 4 \end{array}$$

$$2y^2 \begin{cases} 0 & 0 & 1 & 4 \\ 2 & 2 & 3 & 6 \end{cases} \rightarrow \text{No } 5$$

3. $(x^2 + y^2)^2 - 2(3x^2 - 5y^2)^2 = z^2$ no integral solution.

$$0^2 \equiv 2^2 \equiv 0 \pmod{4}$$

$$1^2 \equiv 3^2 \equiv 1 \pmod{4}$$

either 0, 1

$$(x, y) = (\text{odd}, \text{even}) \Rightarrow (1)^2 - 2(3)^2 \equiv -1 \pmod{4} \rightarrow \leftarrow$$

$$= (\text{even}, \text{odd}) \Rightarrow (1)^2 - 2(-5)^2 \equiv -1 \pmod{4} \rightarrow \leftarrow$$

either 0, 1

case ① x, y both even

$$\begin{aligned} \text{let } x_0 &= 2x_1 \\ y_0 &= 2y_1 \end{aligned}$$

$$\begin{aligned} z_0^2 &= ((2x_1)^2 + (2y_1)^2)^2 - 2(3(2x_1)^2 - 5(2y_1)^2)^2 \\ &= 16 \{ (x_1^2 + y_1^2)^2 - 2(3x_1^2 - 5y_1^2)^2 \} \end{aligned}$$

$$z_1^2 = \{ (x_1^2 + y_1^2)^2 - 2(3x_1^2 - 5y_1^2)^2 \} \rightarrow \leftarrow$$

$$z_0 = 4z_1$$

case ② x, y both odd

$$1^2 \equiv 3^2 \equiv 5^2 \equiv 7^2 \equiv 1 \pmod{8}$$

$$\Rightarrow (x^2 + y^2)^2 \equiv 4 \pmod{8}$$

$$\Rightarrow (3x^2 - 5y^2)^2 \equiv 4 \pmod{8}$$

$$\Rightarrow \text{LHS} \equiv 12 \pmod{8}$$

$$\begin{array}{ccc} & 4^2 & 6^2 \\ \hline 0^2, & 2^2 & 1^2 \equiv 3^2 \equiv 5^2 \equiv 7^2 \\ \hline 0 & 4 & 1 \\ \hline \text{---} & \text{---} & \text{---} \\ 12 & 12 & 12 \end{array}$$

$$4. \quad x^2 + y^2 + z^2 = 2xyz \quad \text{if} \quad x=y=z=0$$

$$(x, y, z \in \mathbb{Z})$$

case ① x_0 even

y, z odd

$$2x_1^2 + y^2 + z^2 = 4x_1 y z$$

$$\underbrace{x_1^2}_{\text{even}} + \underbrace{y^2 + z^2}_{\text{even}} = \underbrace{2x_1 y z}_{\text{even}}$$

hence x_1 even ↻
무한히 반복

→ ←

case ② all even

$$\cancel{4}(x_1^2 + y_1^2 + z_1^2) = \cancel{16} x_1 y_1 z_1$$

$$\left(\frac{x_1}{2}\right)^2 \equiv 1, \quad \left(\frac{y_1}{2}\right)^2 \equiv 0 \pmod{4}$$

$$1 + 0 + 0 \not\equiv 0 \pmod{4}$$

$$1 + 1 + 0 \not\equiv 0 \pmod{4}$$

$$1 + 1 + 1 \not\equiv 0 \pmod{4}$$

→ x_1, y_1, z_1 는 모두 짝수

$$\Rightarrow \cancel{4}(x_2^2 + y_2^2 + z_2^2) = \cancel{32} x_2 y_2 z_2$$

$$x_1 = 2x_2$$

$$y_1 = 2y_2$$

$$z_1 = 2z_2$$

$$\equiv 0 \pmod{4}$$

.... x y z 는 2로 무한히 나눌 수 있음.

$$\Rightarrow x=0$$

$$y=0$$

$$z=0$$

$$3. \quad 3x^2 + 5y^2 + 7z^2 + 9xy + 11yz + 13zx = 0$$

$$x=1, y=1, z=-1$$

moreover

$$\text{then} \quad \underbrace{3+5+7+9}_{24} - \underbrace{11+13}_{-24} = 0$$

$$x=k$$

$$y=k$$

$z=-k$ 는 모든 해가 됨.